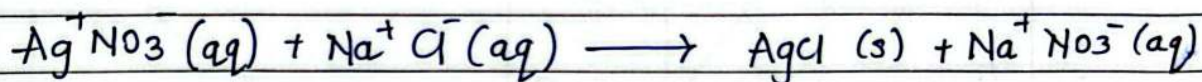


G~3

Chemical Kinetics.

⇒ Slow and Fast Reactions



This reaction occurs between ions and products are obtained as soon as reactants are mixed.

————→ These are fast reactions or ionic reactions.

Rusting of iron which can take months and are very slow.

————→ These are very slow reactions.

⇒ The branch of chemistry which deals with the study of reaction rates and mechanism by which they occur is called Chemical Kinetics.

⇒ The rate of a chemical reaction refers to the speed which reactants change into products.

⇒ The rate of a chemical reaction may be defined as the change in concentration of any of the reactants or any of the products per unit time.

$$\text{Rate of reaction} = \frac{\left[\begin{array}{c} \text{Change in concentration} \\ \text{of reactant / product} \end{array} \right]}{\text{Time taken for change.}}$$

⇒ Average Rate of Reaction.

The change in concentration of any of the reactants or any of the products per unit time over a specified interval of time is called the average rate of the reaction.

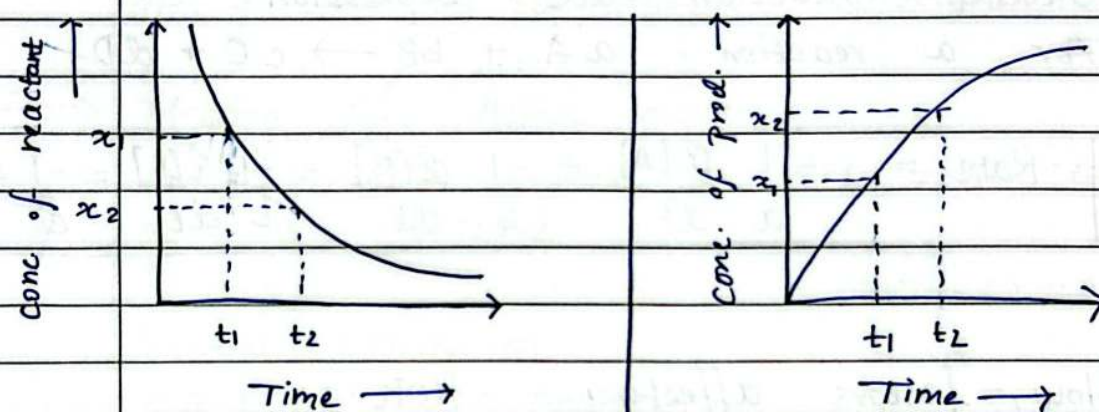
⇒ Instantaneous rate of Reaction.

The rate of change of concentration of any of the reactants or products at a particular instant of time is called instantaneous ^{rate of} reaction at that instant of time.

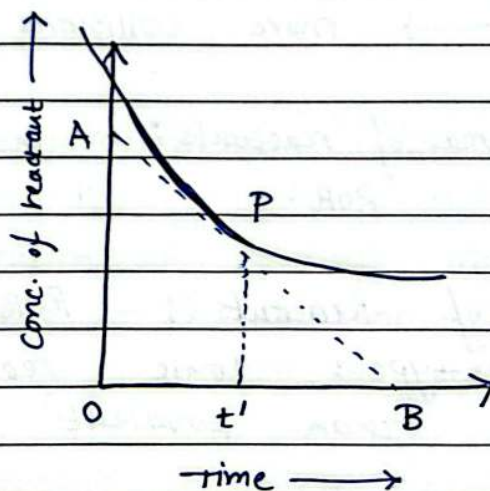
i]
$$\text{Average rate} = \frac{\Delta[\text{conc}]}{\Delta t} = \frac{C_2 - C_1}{t_2 - t_1}$$

ii]
$$\text{Instantaneous rate} = -\frac{d[R]}{dt} \longrightarrow \text{Reactant.}$$

$\Rightarrow$ Graphical Representation.



[AVERAGE RATE OF REACTⁿ]



[Instantaneous Rate of Reaction]

⇒ Factors affecting rate of reaction.

$$\text{Rate of Reaction} = \frac{-\Delta[R]}{\Delta t} = + \frac{\Delta[P]}{\Delta t}$$

$$R = A$$

$$P = B$$

⇒ General reaction rate expression.

For a reaction: $aA + bB \rightarrow cC + dD$.

$$\text{Rate} = -\frac{1}{a} \frac{d[A]}{dt} = -\frac{1}{b} \frac{d[B]}{dt} = -\frac{1}{c} \frac{d[C]}{dt} = -\frac{1}{d} \frac{d[D]}{dt}$$

Now, factors affecting ROR.

i] Concentration of reactants & ROR $\propto$ Concentration (generally). More molecules per unit volume $\rightarrow$ more collisions $\rightarrow$ more rate.

ii] Surface area of reactants: Increased surface area increases ROR.

iii] Nature of Reactants: ROR for gases > liquid > solids
Bond type: Ionic reactions are much faster than covalent.

iv] Temperature: ROR increases with temperature. roughly doubles for every 10°C .

v] catalyst / Radiation: Both increase and accelerate reactions.

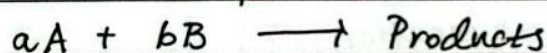
$\Rightarrow$ Order and Molecularity of a Reaction.

1. Law of Mass Action: It states that the rate of a chemical reaction at a given temperature is directly proportional to the product of active masses of reacting substances each raised to a power of their stoichiometric constant.

$\Rightarrow$ Meaning of Active mass

Active mass refers to the molar concentration of a reactant

General explanation



$$\text{Rate} \propto [A]^a [B]^b$$

$$\text{Rate} = K [A]^a [B]^b$$

$\Rightarrow$ Effect of concentration on Rate of Reaction

i] As concentration of reactants increases active mass increases $\rightarrow$ collision frequency increases $\rightarrow$ rate of reaction increases.

ii] This qualitative dependence combined with law of mass action leads to formal Rate law.

$\Rightarrow$ Rate law (Rate Equation)

An experimentally determined equation relating the rate of a reaction to concentration of reactants each raised to power.

$$\text{Rate} = k [A]^x [B]^y$$

i) k = rate constant / specific to reaction depends on temperature.

ii) x, y = "orders" with respect to A and B determined experimentally for the overall reaction.

$\Rightarrow$ order of the reaction.

It is equal to the sum of the powers of concentration terms in the experimentally determined rate law.

$$\text{Order} = x + y \quad (\text{from } [k [A]^x [B]^y])$$

$\rightarrow$ It is an experimental quantity and it must be determined from real concentration data.

$\rightarrow$ It can match stoichiometric constant only for a simple elementary reaction.

i) Zero order $\Rightarrow$ Rate = k

ii) 1st order $\Rightarrow$ Rate = $k [A]^1$

iii) 2nd order $\Rightarrow$ Rate = $k [A]^2$

⇒ Integrated Rate Equations and half life
(zero order and first order only)

Integrated rate equations express concentration as a function of time, obtained by integrating the rate law.

$$\text{Rate} = k[A]^0$$

i) Formula for zero order :

$$k = \frac{A_0 - A_t}{t}$$

$$t_{1/2} = \frac{[A_0]}{2k}$$

ii) Formula for first order :

$$k = \frac{2.303}{t} \log \left(\frac{A_0}{A_t} \right)$$

also exponential form : $[A]_t = [A]_0 e^{-(kt)}$

$$t_{1/2} = \frac{0.693}{k}$$

Note : A_0 is initial concentration at $(t=0)$
 A / A_t is concentration at 't'

$\Rightarrow$ Pseudo first Order Reactions

The reactions which apparently seem to be of higher order but under circumstances behave as first order reaction.

$\Rightarrow$ Molecularity of the Reaction.

Meaning - Physical nature.

- Molecularity = number of reactant molecules (atoms/ions) that must collide simultaneously in a (single-step) reaction to bring about the chemical change.

Unimolecular - one molecule - decomposing / isomerising.

bimolecular - two molecule colliding.

termolecular - three molecules (rare).

It is always a whole number, never 0 or fraction.

For a simple reaction, order = molecularity.

$\Rightarrow$ concept of Energy.

i) Exothermic Reactions: Energy of products is lower than energy of reactants. Energy is released (ΔH is -ve).

ii) Endothermic Reactions: Energy of products is higher than energy of reactants. Energy is absorbed. (ΔH is +ve)

=> Energy Barrier

Even though a reaction may be thermodynamically favourable (exothermic), reactant molecules must still cross an energy barrier or hill, before converting to products. This is represented by a peak on potential energy vs coordinate diagram.

=> Threshold Energy (E_t)

The minimum total energy that colliding molecules must possess for a collision.

=> Activation Energy (E_a)

The additional energy that must be supplied to reactant molecules to raise them up to the threshold energy so the reaction can occur.

$$E_a = E_{(\text{threshold})} - E_{(\text{average reactants})}$$

=> All molecules pass through a high-energy transient state called activated complex

$$\Delta H = E_a - E_b.$$

Note: A catalyst provides an alternate reaction pathway with a lower activation energy

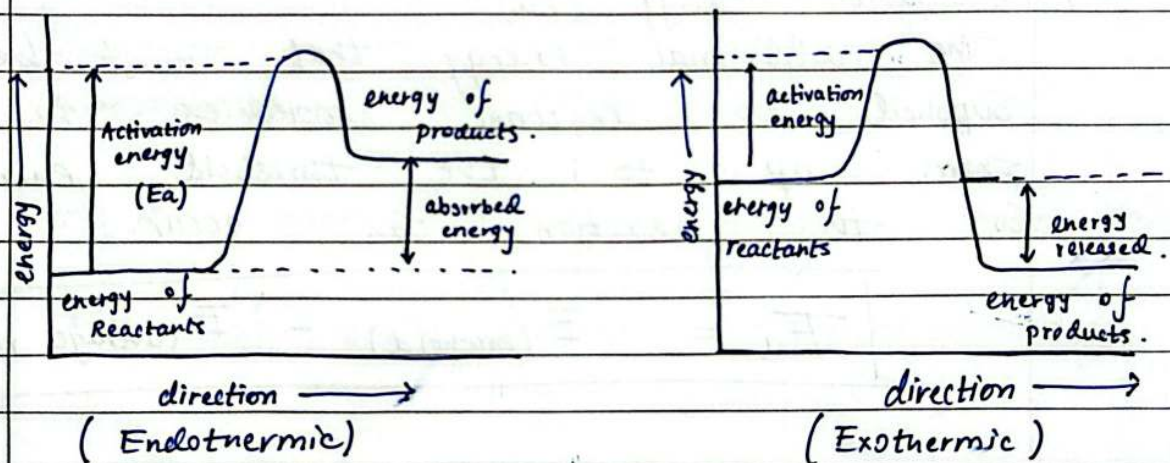
⇒ Collision Theory

Condition for a chemical change

i) Reactant particles must come into close contact - they must collide with one another for any chance of reaction.

→ optimum / sufficient Energy.
→ Proper orientation.

Energy Diagrams (Exothermic and Endothermic)



⇒ Mechanism of Elementary Reaction.

It is a reaction which occurs in a single step.

⇒ Mechanism of complex Reaction.

A complex reaction proceeds through two or more elementary steps.

⇒ Effect of Temperature on Rate constant

$$K = A e^{\frac{-E_a}{RT}}$$

rate constant $\rightarrow$ A (Arrhenius factor) $e^{\frac{-E_a}{RT}}$ (Temperature) E_a (Activation energy) R (Gas constant)

$$\log K = \log \left(\frac{A - E_a}{2.303 RT} \right)$$

for two comparison.

$$\left[\log \left(\frac{K_2}{K_1} \right) = \frac{E_a}{2.303 R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right) \right]$$

~~Graph of log K vs 1/T~~

$$\text{Slope} = - \frac{E_a}{2.303 R}$$

$$\boxed{\text{Slope (m)} = - \frac{E_a}{2.303 R}}$$

⇒ Types of Numericals.

1. Order rate law, and units of k .
2. Zero Order Kinetics.
3. First Order Kinetics.
4. Arrhenius Equation.
5. Reaction rate expression (rare).

Master Units Table.

Rate of Reaction	$\text{mol L}^{-1} \text{s}^{-1}$
k (zero order)	$\text{mol L}^{-1} \text{s}^{-1}$
k (first order)	s^{-1}
k (second order)	$\text{L mol}^{-1} \text{s}^{-1}$
k (nth order)	$\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}$
Half life ($t_{1/2}$)	s (or min).
Activation Energy (E_a)	J mol^{-1}

Types.

1) Order, Rate law and k .

→ Type 1: Given the rate law, find order.
 $\text{Rate} = k [\text{A}]^2 [\text{B}] \rightarrow \text{order} = 2+1 = 3.$

→ Type 2: Given the units of k find the order

for n^{th} order: $[\text{mol}^{1-n} \text{L}^{n-1} \text{s}^{-1}]$.

Eg:- $(\text{mol L}^{-1} \text{s}^{-1})$ is of zero order.

2) zero Order kinetics.

Formula :

$$k = \frac{A_0 - A}{t}$$

and

$$t_{1/2} = \frac{A_0}{2k}$$

→ Type 1) Find k , given $[A_0]$, $[A]$ and $[t]$.

→ Type 2) Find $[A]$, given $[A_0]$, t and k .

→ Type 3) Find time for a given amount to react.

Eg:- "time for concentration drop from 0.5 M to 0.2 M" → $t = \frac{A_0 - A_t}{k}$

If instead it says 80% completed then
 $A = 0.2 [A_0]$ (20% remains).

3) First order kinetics (Commonly asked).

Formula :

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}$$

and

$$t_{1/2} = \frac{0.693}{k}$$

→ Type 1) Find k , given $[A_0]$, $[A]$ and t

→ Type 2) Find $[A]$ or t when data is given as % completion

• 50% completed → $[A] = 100 - 50 = 50$.

$$100/50 \Rightarrow A_0/A = 2.$$

• 75% completed → $[A] = 100 - 75 = 25$.

$$100/25 \Rightarrow A_0/A = 4.$$

• 90% completed → $[A] = 100 - 90 = 10$.

$$100/10 \Rightarrow 100/10 = 10.$$

• 99% completed → $[A] = 100 - 99 = 1$.

$$\Rightarrow 100/1 = 100.$$

→ Type 3) Half life.

$$t_{1/2} = 0.693/k.$$

direct substitution questions.

→ Type 4) Time for conc. to fall to a specific fraction using half lives directly

If asked "time for concentration to become 1/4 of initial" → $2 \times \text{half lives} = 2 \times t_{1/2}$.

→ Type 5) Craseous reactions.

Just replace $[A_0]$ with P_0
and $[A]$ with P

4) Arrhenius Equation

→ Type 1: find E_a given k, A, T .

$$\log k = \log A - \frac{E_a}{2.303RT}.$$

→ Type 2: (much common) two temperatures.

$$\log \frac{k_2}{k_1} = \frac{E_a}{2.303R} \left(\frac{1}{T_1} - \frac{1}{T_2} \right).$$

→ Type 3: Graphical form.

"Given a graph of $\log k$ v/s $1/T$.
find E_a from slope."

$$\text{Slope} = - \frac{E_a}{2.303R}$$

$$\text{Intercept} = \log A.$$

5) Rate Equation

Directly form rate Equation from.

$$\frac{-1}{a} \frac{d[A]}{dt}$$

$$\frac{-1}{b} \frac{d[B]}{dt}$$

$$\frac{-1}{c} \frac{d[C]}{dt}$$

$$\frac{-1}{d} \frac{d[D]}{dt}$$

6) Tables ROR and find order.

→ Data for concentration of Reactants and rate of reaction (k) is given find orders of [A] and [B].